

Practicum No.9**Date: 21/02/2026****Lean Canvas Model Development for a CSE/AIML/DSCS/DS Venture****I. Practical Significance**

Developing a Lean Canvas Model for a CSE/AIML/DSCS/DS venture enables students to translate conceptual ideas into a structured, actionable business framework. It helps learners identify the core elements of their technology-driven start-up—such as the problem, customer segments, unique value proposition, solution architecture, cost structure, and revenue streams—within a single visual page. This exercise enhances strategic thinking by forcing clarity, prioritization, and evidence-based decision-making. For technical students, the Lean Canvas provides a practical pathway to validate technical feasibility, assess market fit, and align product development with real customer needs. Ultimately, this hands-on approach prepares learners to design scalable, AI-enabled, and data-driven ventures with strong entrepreneurial foundations.

II. Competency and Industry-Specific Practical Skills

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a Pitch deck, project proposals for small/medium-scale startups logically and scientifically.
- a) **Interpret** competitive dynamics shaping CSE-focused high-tech and software-driven markets.(CDO).
- b) **Evaluate** commercial viability of CSE products such as AI models, software applications, analytics platforms, and cloud-native solutions.
- c) **Conceptualize** high-impact CSE product opportunities in AI-driven platforms, autonomous systems, digital twins, and next-generation software solutions.
- d) **Practice** professional ethics in data collection, system design, and user interaction within digital ecosystems.

III. Relevant CO(s)

- CO3: **Formulate** a startup proposal for the chosen idea through entrepreneurial support systems.

IV. Practical Outcome (PrO)

- **Present** a business model for **the chosen** CSE, AIML, DSCS & DS start-up using the 9-block Lean Canvas Model.

V. Related ADO(s) (keep as such)

- a) **Practice** professional ethics.
- b) **Function** effectively as a team member/leader.
- c) **Practice** integrity and honesty.

VI. Minimum Underpinning Theory

The development of a Lean Canvas model for a CSE/AIML/DSCS/DS venture is grounded in core entrepreneurial and computing-innovation theories that explain how technology-driven startups identify value, validate solutions, and build scalable business models. The Lean Startup methodology forms the foundation, emphasizing rapid experimentation, customer discovery, iterative prototyping, and evidence-based decision-making. Complementing this are key concepts from software-product innovation, including problem-solution fit, product-market fit, and the dynamics of competitive advantage in AI, data science, cloud-native systems, and cybersecurity domains. Market segmentation theory, technology adoption models, and value proposition design further support the structured analysis of customer needs, early adopters, and measurable benefits. Ethical frameworks in computing—covering responsible data use, system reliability, and user-centered design—provide essential guidance for ensuring that the proposed venture aligns with professional standards and societal expectations. Together, these theories underpin the design of a robust and realistic Lean Canvas tailored for high-tech digital and intelligent systems ventures.

VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

VIII. Resources Required

S. No.	Name of Resource	Suggested Broad Specification	Quantity
1	Laptop/PC	For research and documentation	1 per team
2	Internet access	Trend research	As required
3	Brainstorming sheets/chart paper	For idea listing	1 set
4	Reference materials	Magazines, journals, websites	1 per team

IX. Safety Precautions

- a) **Avoid** plagiarism.
- b) **Maintain** originality in problem identification.
- c) **Respect** privacy and data ethics.
- d) **Ensure** solution feasibility.

X. Procedure

1. **Form** a team and select a startup idea.
2. **Identify** the customer segments and core problems.
3. **Brainstorm** the value proposition and solution.
4. **Define** key metrics, channels, and revenue model.
5. **Identify** cost structure and key partners.
6. **Document** the Lean Canvas in the recommended format.
7. **Submit** for evaluation.

XI. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1	Laptop	ASUS	Used for research, Lean Canvas preparation, and documentation	1	Laptop
2	Internet Access	Wi-Fi	Used to access Lean Canvas templates and startup references	As required	Internet Access
3	Notepad	Classmate	Used for brainstorming and listing startup ideas	1	Notepad
4	Reference Materials	Journals	Used for studying Lean Canvas model and startup examples	1	Reference Materials

XII. Actual Procedure Followed

1. Selected the mini project idea **Digital Attendance and Productivity Tracker** as the startup concept.
2. Studied the **Lean Canvas business model framework** and understood its nine components.
3. Identified the main **problem**, which is the difficulty teachers and institutions face in managing attendance and monitoring student productivity manually.
4. Determined the **target customer segments**, including educational institutions, teachers, and students who need a digital attendance tracking system.
5. Designed a **technology-based solution** that automatically records attendance and tracks productivity using a web or mobile application.
6. Completed all sections of the **Lean Canvas model**, including problem, customer segments, value proposition, solution, channels, revenue streams, cost structure, key metrics, and unfair advantage.
7. Reviewed and refined the Lean Canvas to ensure **technical feasibility, market relevance, and practicality**.

XIII. Observations and Data

S. No.	Venture Identified	Data/ Remarks
1	Problem Identification	Manual attendance tracking is time-consuming and inefficient
2	Customer Segments	Educational institutions, teachers, and students
3	Value Proposition	Automated attendance and productivity monitoring system
4	Solution Offered	Web or mobile platform for attendance tracking and performance insights
5	Revenue Model	Subscription-based service for institutions and premium features

XIV. Results/Observations

1. The Lean Canvas model clearly defines the business structure of the Digital Attendance and Productivity Tracker startup idea.
2. The product provides value by reducing manual attendance work and offering productivity insights for students and teachers.
3. The analysis shows that the system has potential demand in schools, colleges, and training institutions.
- 4.

XV. Interpretation of Results

The Lean Canvas framework helps in understanding the entire startup model in a clear and structured way. It helps identify the main problem, define the target users, and plan how the product will generate value and revenue. For the **Digital Attendance and Productivity Tracker**, the Lean Canvas shows that the idea is practical and aligns with the growing use of digital tools in education.

XVI. Conclusions

The Lean Canvas analysis shows that the **Digital Attendance and Productivity Tracker** is a feasible startup idea with a clear problem–solution fit. The system has potential to improve attendance management and academic productivity in educational institutions. Using the Lean Canvas model also helped in organizing the business idea and identifying improvements needed for successful implementation.

XVII. Suggested Practicum Related Questions

1. **Analyze** how customer segment identification strengthens the Lean Canvas of a CSE/AIML/DSCS/DS venture.
2. **Evaluate** how unique value propositions differentiate AI/ML and cybersecurity solutions in competitive markets.
3. **Assess** how cost structures (cloud, compute, data pipelines, licensing) affect financial viability in the Lean Canvas.
4. **Examine** how competitive threats and emerging technologies shape the “Unfair Advantage” of computing ventures.
5. **Interpret** how Lean Canvas insights influence MVP development, customer validation, and product-market fit.

XVIII. References / Suggestions for further reading (keep as such)

- a) Venture Meets Mission: Aligning People, Purpose, and Profit to Innovate and Transform Society, Arun Gupta, Gerard George, Thomas J. Fewer, Stanford University Press, Stanford, California, 2024, ISBN: 978-1503636283
- b) So You Want to Start a Business: Eight Steps to Startup Success, John B. Vinturella, Business Expert Press, New York, USA, 2024, ISBN: 978-1637425855
- c) Business Planning, Franklin A. Gevurtz, Foundation Press (West Academic), St. Paul, Minnesota, 2024, ISBN: 978-1685612764

XIX. Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the related competency of the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related: 18 Marks - 60%			
1	Identification of domain & problems	3	
2	Use of structured idea generation techniques	4	
3	Team collaboration & creativity	4	
4	Feasibility evaluation	3	
5	Ethical originality & clarity	4	
Product-related Skills: 12 Marks - 40%			
1	Quality & novelty of ideas	2	
2	Logical presentation & organization	2	
3	Clarity of findings & conclusions	3	
4	Report/presentation submission	3	
5	Innovation potential & market relevance	2	
Total		30	

To be filled by Student Team Leader:

S.No.	Name of the Student	Register No.
1	Neavis Rishva. J	URK25CS1236

To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Signature with Date
Process-Related skills (18)	Product-Related skills (12)	Total (30)		

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